

IMAGINARY ELECTRIC COMPANY

Power Generation Division - Plant Integration

Alpilla 600 MW Combined Cycle Power Plant - 1 x IE-9H.02 multi-shaft power island

Gas Turbine IE-9H.02 - Datasheet

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	Prepared	Checked	Approved
Function	IEC plant integration engineer	IEC lead engineer (discipline)	IEC project engineering manager
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CASE STUDY - Imaginary Electric, its products and all data are fictional.

1. General

Item	Data
Model	IE-9H.02 heavy-duty single-shaft gas turbine, 50 Hz, 3,000 rpm, cold-end drive, axial exhaust
Fleet experience	11 units in commercial operation, lead unit 38,000 fired hours (ER-01.08: complies)
Compressor	14 stages, pressure ratio 23.8:1, variable IGV + 3 rows of variable stator vanes
Combustion system	IE DLN-H2, 16 can-annular combustors, dual fuel (natural gas / LDO), water injection on LDO, up to 30 vol % H2 capable hardware (5 % without modification)
Turbine	4 stages, air-cooled, single-crystal stage 1 blades, thermal barrier coatings
Starting	Static frequency converter (SFC) through the generator, 8.5 MW input from 10.5 kV
Enclosure	Acoustic enclosure 85 dB(A) at 1 m, ventilation 2 x 100 % fans, gas detection, CO2 fire suppression
Inlet air system	Filter house 3 stages: G4 weather louver/coalescer, F9, EPA E12; anti-icing by compressor bleed; silencer. Evaporative cooler: space provision only, not included

2. Ratings

Condition	Output MW	Heat rate kJ/kWh	Efficiency %	Exhaust kg/s	Exhaust degC
ISO 2314 (15 degC, 60 %, 1013.25 mbar, no inlet/outlet losses), gas	432.0	8,240	43.69	801.0	638.0
SRC-NG-100 (10/36 mbar losses, fuel 215 degC)	425.0	8,314	43.30	796.7	640.0
SRC, LDO with water injection (base load)	395.0	8,650	41.62	812.0	622.0
Maximum output (cold ambient, generator/shaft limit)	470.0	-	-	-	-

3. Fuel requirements (at IEC terminal points, see IEC-ALP-IF-001)

Parameter	Natural gas (IF-01, performance gas heater inlet)	LDO (IF-02, liquid fuel skid inlet)
Pressure	36.5 barg minimum at base load; 41.0 barg maximum; max. rate of change 0.7 bar/s	3.0 - 6.0 barg
Temperature	5 - 40 degC and at least 28 K above the hydrocarbon and water dew points	10 - 45 degC
Flow at base load (SRC)	20.67 kg/s	22.1 kg/s (+ demin water 22.5 kg/s at IF-03)
Quality	ALP-OWN-FUL-001 range; Wobbe index change <= 1 %/min; trace metals Na+K <= 0.02 mg/Sm3 equivalent; particulates filtered to 3 um absolute (EPC final filter)	ER-03.07 limits at the GT inlet; filtration 5 um (IEC skid)
Fuel heating	IEC performance gas heater to 215 degC with IP feedwater; start-up heating by EPC gas station (electric or water-bath heater, gas >= 28 K superheat)	-

4. Emissions and noise

Parameter	Natural gas	LDO
NOx at GT outlet (15 % O2, dry), 40-100 % load	<= 50 mg/Nm3	<= 90 mg/Nm3 (water/fuel ratio 1.02)
CO at GT outlet, 40-100 % load	<= 20 mg/Nm3	<= 25 mg/Nm3
Minimum emissions-compliant GT load (MECL)	40 % GT load (with SCR)	50 % GT load
Near-field noise, enclosure and inlet	85 dB(A) at 1 m	85 dB(A) at 1 m

5. Operation and maintenance

Item	Data
Start to base load (GT)	hot start 25 min, warm 35 min, cold 45 min (GT only; CC limited by HRSG/ST)
Load ramp (GT)	40 MW/min normal, 65 MW/min fast
Combustion inspection / hot gas path / major inspection	12,500 / 25,000 / 50,000 EOH, or 900 / 900 / 1,800 starts, whichever first
Equivalent operating hours (EOH)	EOH = fired hours + 20 x starts + 60 x trips from base load + 2 x h on LDO
Lube oil	ISO VG 32 mineral, tank 28 m3, 2 x 100 % AC pumps + DC emergency pump; cooling duty 1.6 MW
Compressor washing	on-line and off-line; detergent skid by IEC; demin water 5 m3 per off-line wash

6. Dimensions, weights and transport

Item	Dimensions L x W x H (m)	Weight (t)
GT core engine (shipping, with transport frame)	13.2 x 5.9 x 5.6	405
Exhaust diffuser (2 pieces)	8.5 x 7.2 x 7.0	2 x 62
Inlet filter house (modules)	18 modules	max. 28 per module
GT rotor (maintenance lift)	11.0 x 3.2 x 3.2	112

The GT core engine exceeds the 120 t road limit (ALP-OWN-GEN-001): delivery by barge to the landing TP-M1; the EPC shall design the barge landing and the haul road for a 480 t gross transport (SPMT).

7. Auxiliary power (base load, SRC)

GT package auxiliaries 1,180 kW (lube oil, hydraulic, enclosure ventilation, fuel gas skid, filter house); SFC 8.5 MW during start-up only. See IEC-ALP-REQ-001 table 7.1.